

# Project 1 - Data Warehouse Design

## Data Warehousing (CITS5504)

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# 1. Introduction

Reducing the number of deaths caused by car crashes is a key concern for both governments and the general public [1]. Members of the United Nations Road Safety Collaboration (UNRSC) addressed this issue during the Decade of Action for Road Safety, implementing a range of initiatives aimed at stabilising and reducing road traffic fatalities.

Addressing this issue is the focus of the present analysis, which is considered of extreme global importance. The approach proposed in this paper is the implementation data warehouse concept to facilitate the analysis and act as a supplement for Business Intelligence (BI) tools.

The methodology for this analysis is the organisation of the historical data of fatal crashes from the Australia Road Death Database [2], into a data warehouse data model for Online Analytical Processing (OLAP). The aim of this OLAP system is to be able to analyse the data for decision making, through the ability of presenting the data in various formats and aggregations. The present project follows the Four-step dimensional design process proposed by Kimball and Ross, to identify fact table elements, and the corresponding dimensions [3]. The design principle of this model is to define business questions that will allow drawing conclusions and identifying frequent patterns in road fatalities. As a result, this study will provide recommendations aimed at minimising the incidence of the factors that are most commonly found in fatal traffic incidents. For the design of this data warehouse, the dimensions are modeled in a snowflake schema, with the majority of the dimensions connecting to the fact table, and one dimension providing additional details to another dimension (Population to Location).

In this project, the formulated questions are presented in a star-net format, where the hierarchy of the data warehouse dimensions are represented. From the star-net, the corresponding queries can be formulated to obtain the necessary data and aggregation to the level required by the business question.

The Extraction Transformation and Loading (ETL) of the data is centered on the fatalities data set, which records features such as location, time, date, demographic information of the road user at a human casualty grain level. The data is then transformed and cleaned to resolve missing values and loaded into the database server for querying and retrieval from visualisation tools. In addition, GeoJSON geographical data corresponding with LGA codes was loaded to generate map visualisations from Tableau.

The implementation of this data warehouse is carried out on PostgreSQL, administered and managed on pgAdmin, a tool that provides database server administration and the interface for elaborating and managing SQL queries. For visualisation, the Tableau tool was employed, leveraging the connection capability to PostgreSQL, and accessing the GeoJSON geographical data for map visualisations.

Finally, the Association Rule Mining methodology was employed. This is a powerful tool that identifies patterns and features that are frequently found together in the data set. This tool provides insights on probabilities such as fatalities occurring for specific road user demographics, given a set of repeating circumstances, such as road conditions or day of the week.

## 2. Body

### 2.1. Business Questions

The purpose of this data warehouse is to be able to provide answers to business questions, leading to the identification of patterns or associations, which determine the significant probability of a fatality. More specifically, the design of this data warehouse will be modelled to be able to respond to the questions listed below, or of similar nature.

1. What is the number of fatalities at different times of the day?
2. What is the number of crashes at night from 2020 of every states?
3. What is the number of fatalities on holiday at every speed limit?
4. What is the percentage of each type of crash by gender?
5. What is the number of elderly fatalities in every LGA in WA state?
6. What is the average monthly number of fatalities of young driver in the *Brisbane* LGA?
7. What is the ratio of fatalities over population in Perth in 2024?

To be able to respond to these questions, our data needs to be structured and stored following the four steps proposed by the Kimball's dimensional design model [3]:

- Identify the process being modelled
- Determine the grain at which facts can be stored
- Choose the dimensions
- Identify the measures for the facts

### 2.2. Data Warehouse Design

As introduced in the previous section, we adopt the Four-step Dimensional Design Process to define the data warehouse design.

#### **Select The Business Process**

The business process we are interested in, are the fatalities resulting from car crashes. We are identifying these from their common characteristics, described as:

- Processes are identified by verbs, in this case the unfortunate driving leading to a fatal crash. For every incident, there is descriptive data collected into the dimensions associated with the event.
- The operational systems supporting the business process are the emergency services provided by the governments including Police, Fire (metropolitan and rural), and Ambulance services [4]
- The process also generates metrics, like speed-limits, age of the road user, day of the week, road type, location, among others, which are captured in the data set.
- The process of an accident also triggers inputs that become the fact tables that we are modelling in this exercise.

## Declare the Grain

The grain applies to the level of detail that a row in the fact table represents. In this project we are investigating car crash fatalities in Australia, so the grain level represented by every row in our case, is a fatality resulting from a crash. More specifically, the grain in business terms is: *One row per fatality individual from road transport crash, as identified by the states and territories of Australia.* Since a crash can result in multiple fatalities, a fatality belongs to a specific crash. Therefore, the atomic grain of the fact table is the fatality.

## Identify the Dimensions

Dimensions in a data warehouse provide descriptions for every item in the fact table that allow responding to questions of the business process. For our analysis, these dimensions need to provide a high level of details to be able to respond to the questions formulated earlier in this document. For every fatality, the dimensions in our design include:

**Time of day (dim\_time):** allow the analysis to be able to answer questions related to fatalities or crashes that occurred at different times during the day. Although PostgreSQL supports direct link of time, a surrogate key is implemented to facilitate direct relationship of fact and dimensional table in BI tool. The dimension follows a set-grouping hierarchy in a total order.

| Attribute | Type        |
|-----------|-------------|
| time_id   | INT PK      |
| FullTime  | Time        |
| Rush_hour | char        |
| TimeBand  | varchar(20) |

Table 2.2.1. Time dimension schema

| Time Band | Time        |
|-----------|-------------|
| Night     | 00:00-04:59 |
| Morning   | 05:00-11:59 |
| Afternoon | 12:00-17:00 |
| Evening   | 17:00-21:00 |
| Night     | 21:00-23:59 |

Table 2.2.2. Time Band Group Order

**Date (dim\_date):** allows the analysis to answer questions related to fatality occurred in different periods of the year such as day of the week, month, year. The attributes of the dimension follow a total order hierarchy schema: DayWeek -> Month -> Year

| Attribute | Type        |
|-----------|-------------|
| date_id   | Int PK      |
| DayWeek   | Varchar(10) |
| Month     | int         |
| Year      | Int         |

Table 2.2.3. Date dimension schema

**Gender (dim\_gender):** allows the analysis to answer questions related to demographic information of the fatal road user.

| Attribute | Type        |
|-----------|-------------|
| gender_id | Int PK      |
| Gender    | varchar(10) |

Table 2.2.4. Gender dimension schema

**Age (dim\_age):** allows the analysis to answer questions related to the age of the fatal road user. Different age groups can have varied experience and difficulties in traffic participants. This will let the policy makers to target different age groups with more appropriate and efficient strategies. The attributes of the dimension follow a set-grouping hierarchy in a total order.

| Attribute | Type        |
|-----------|-------------|
| Age       | Int PK      |
| AgeBand   | varchar(10) |

Table 2.2.5. Age dimension schema

| AgeBand | Age |
|---------|-----|
|---------|-----|

|        |       |
|--------|-------|
| Young  | <30   |
| Mature | 30-39 |
| MidAge | 40-65 |
| Older  | >65   |

Table 2.2.6. Age Band Group Order

**Road Type (dim\_road):** allows the analysis to learn how different types of road can influence the likelihood of fatality and implement counter measures to tackle that. As the type of road is stored as a long string, surrogate key is implemented to reduce the storage cost.

| Attribute | Type        |
|-----------|-------------|
| road_id   | Int PK      |
| Road_Type | Varchar(30) |

Table 2.2.7. Road Type dimension schema

**Crash type (dim\_crashtype):** allows the analysis to know whether the crash was single or multiple crashes.

| Attribute  | Type        |
|------------|-------------|
| crash_id   | char        |
| Type_Label | Varchar(15) |

Table 2.2.8. Crash dimension schema

**Holiday (dim\_holiday):** allows the analysis to know whether holiday factors like Christmas and Easter have any influence on the fatality. Hence, the authorities can have targeted educational marketing campaigns to reinforce the road users' awareness during the holidays.

| Attribute    | Type        |
|--------------|-------------|
| holiday_id   | int PK      |
| Holiday_Type | varchar(10) |

Table 2.2.9. Holiday dimension schema

**Road user (dim\_roaduser):** allows the analysis to identify the type of fatal user.

| Attribute    | Type   |
|--------------|--------|
| road_user_id | Int PK |

|           |             |
|-----------|-------------|
| User_Type | Varchar(20) |
|-----------|-------------|

Table 2.2.10. Road User dimension schema

**Type of vehicles involved(dim\_involvedtype):** allows the analysis to know if there was and what kind of heavy vehicle was involved in the crash. Each vehicle attribute indicates whether there was any involvement of that type or the information was not recorded

| Attribute   | Type        |
|-------------|-------------|
| Involved_ID | int PK      |
| Bus         | VARCHAR(10) |
| HeavyRigid  | VARCHAR(10) |
| Articulated | VARCHAR(10) |

Table 2.2.11 Involved Type dimension schema

**Speed Limit of the road (dim\_speed):** allows the analysis to see how different speed can influence the crash and fatality. The hierarchy schema of the dimension is a set-grouping schema in a total order.

| Attribute | Type        |
|-----------|-------------|
| SpeedLim  | Int PK      |
| SpeedBand | Varchar(20) |

Table 2.2.12. Speed Limit Dimension

| Speed Band  | Speed  |
|-------------|--------|
| Under50     | <50    |
| From50to100 | 50-100 |
| Over100     | >100   |

Table 2.2.13. Speed Band Group Order

**Location (dim\_location):** allows the analysis to learn the location where the crash happened. This lets the local authority propose and execute the solution for prevention. For this analysis, we will dive into State and Local Government Area (LGA) and try to tailor the implication and demonstrate the capability of the design in drill-down to a lower level. The concept hierarchy of the dimension follows a total order: LGA -> State.

| Attribute | Type        |
|-----------|-------------|
| Loc_ID    | Int PK      |
| LGA_CODE  | Int FK      |
| State     | Varchar(3)  |
| LGA Name  | Varchar(50) |



Table 2.2.14. Location Dimension

**Population (dim\_population):** although traffic accidents can be almost the same wherever it occurs, higher population tends to have more drivers, hence, there are more accidents. The dimension is introduced to provide a more holistic view for policy makers. Analysis can tweak around the ratio of crashes or fatality over population to know which area requires extra attention. The Population is a snowflake dimension connected to the Location dimension. A surrogate key is implemented to differentiate the population at each location each year.

| Attribute     | Type   |
|---------------|--------|
| population_id | int PK |
| location_id   | int FK |
| Year          | Int    |
| Population    | Int    |

Table 2.2.15. Population Dimension Schema

## Identify the Facts

The facts in the dimensional design provide metrics for the grain defined in step 2. The selected facts for this analysis allow the investigation of measures that result in a fatality, allowing the identification of frequent patterns that could be prevented. Regarding the design, a record is corresponding to a fatal road user along with its related details rather than any numerical measure. Therefore, a factless fact table is the most appropriate for the warehouse design.

| Fatalities   | Type       |
|--------------|------------|
| fat_id       | int PK     |
| crash_id     | int        |
| location_id  | intFK      |
| date_id      | intFK      |
| time_id      | intFK      |
| holiday_id   | int FK     |
| crashtype_id | char(1) FK |
| SpeedLim     | int FK     |
| Involved_ID  | int FK     |
| road_id      | Int FK     |
| road_user_id | int FK     |
| Age          | int FK     |

Table 2.2.16. Fatality Fact Schema

## Final Design:

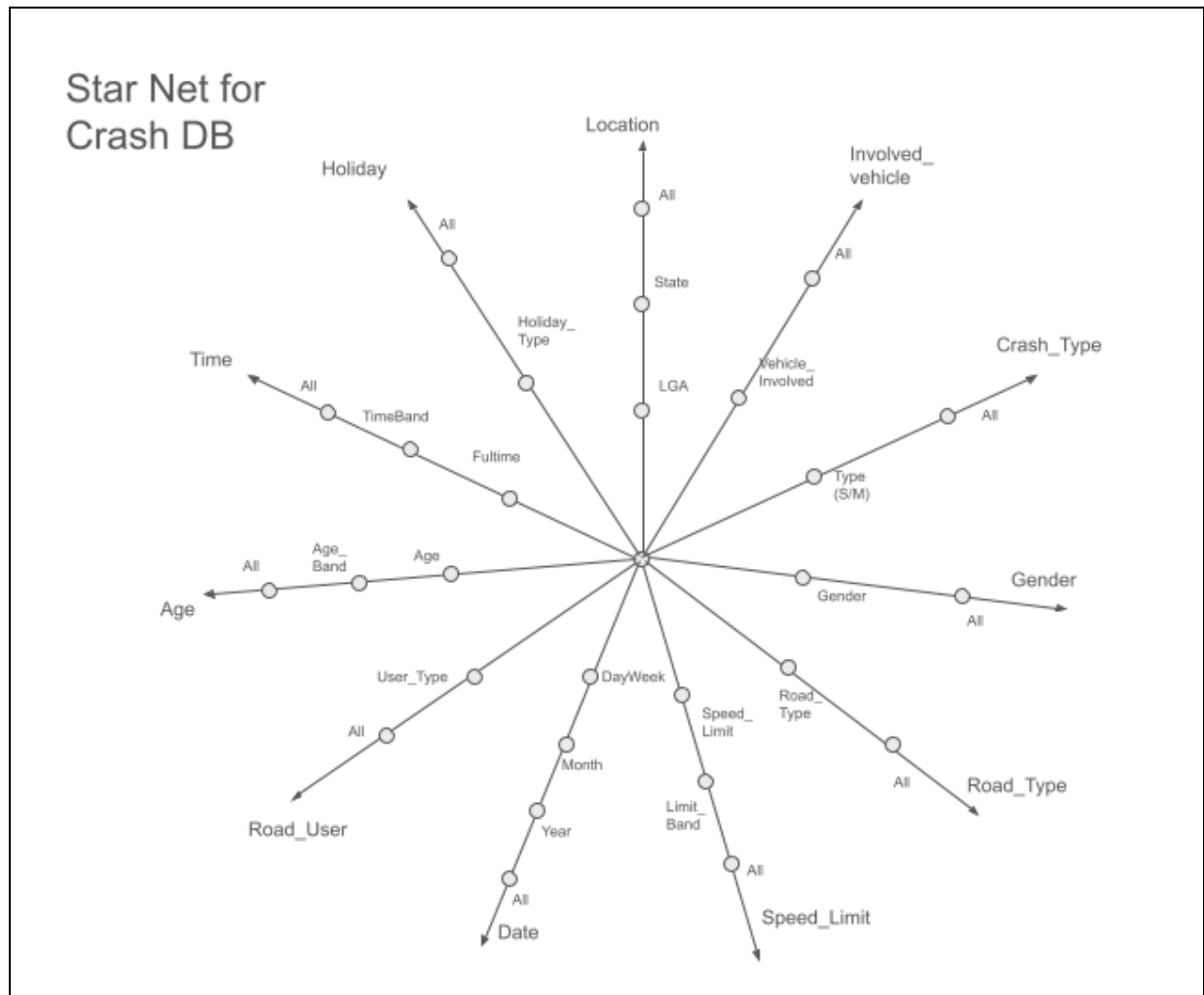
Although the majority of the dimensional tables are directly connected to the fact table, the Population dimension splits from the Location dimension branch. Therefore, in theory, the design aligns with the Snowflake schema where a central fact table contains a bulk of the data and connects to some dimensional tables which may split into additional tables, demonstrated in the picture below [5].



Fig 2.2.17. Data Warehouse Design Schema

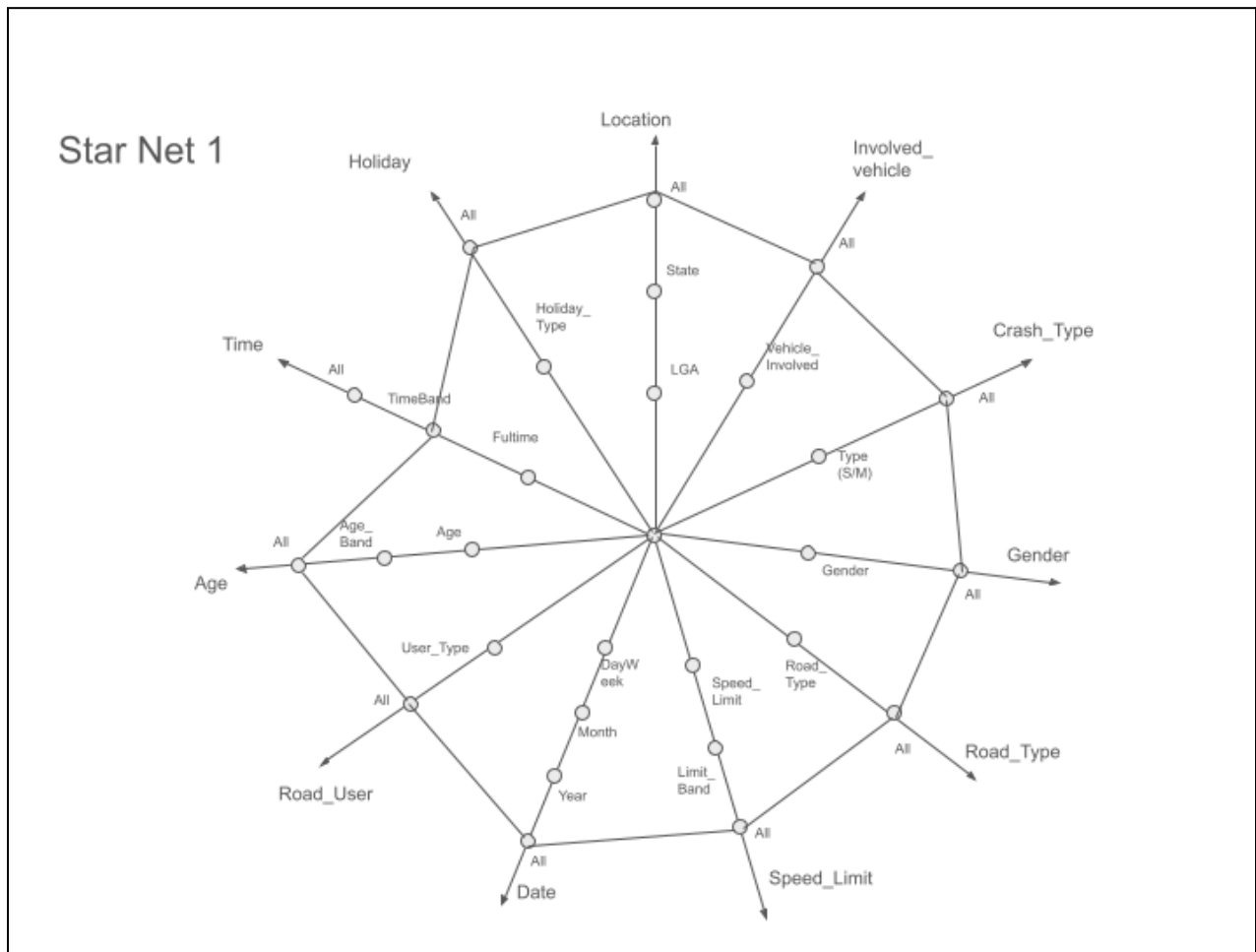
## Star Nets

The dimensions of this data warehouse originate from the business questions formulated for this project. The starnet model represented below, uses radial lines containing the dimension hierarchy, which allows mapping business questions to the data warehouse dimensions.

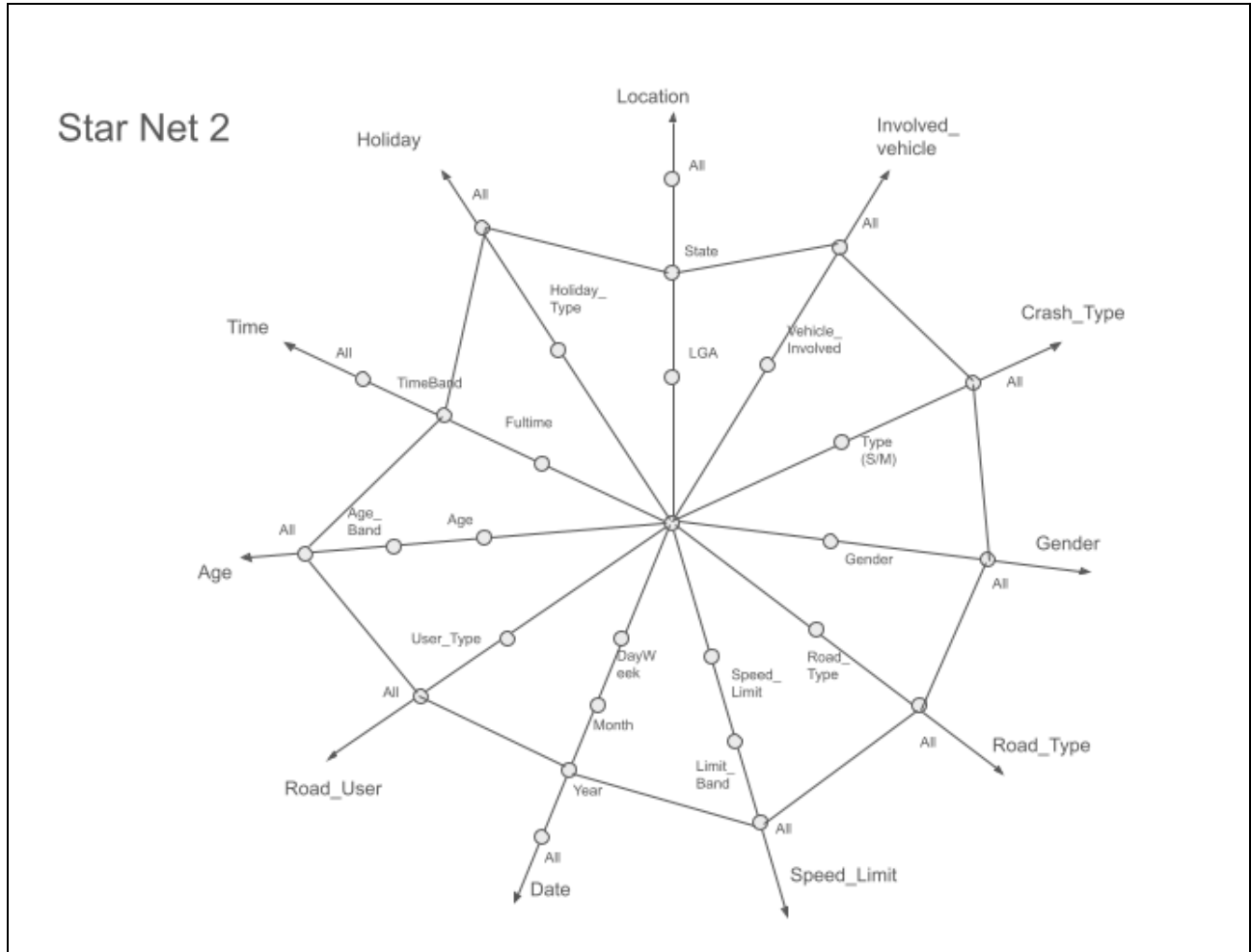


Each abstraction level in the dimension hierarchy is known as a footprint, which identifies at what level the desired data aggregation is performed. The following star net diagrams display the aggregation levels required to respond to the above mentioned questions.

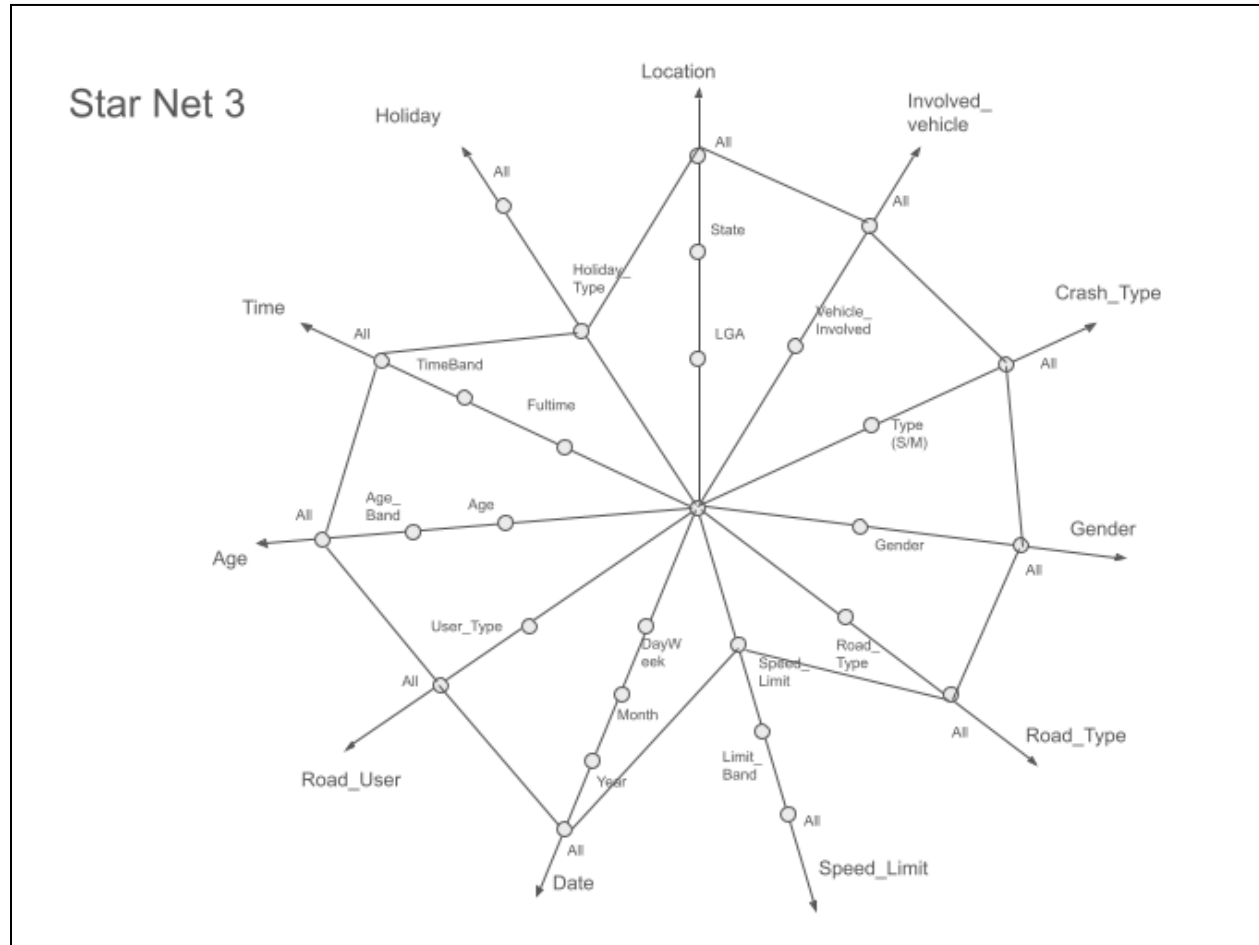
Star Net #1: Number of fatalities at different times of the day. In this star net, our query considers a drill-down to TimeBand in the Time dimension, and the roll-up for all other dimensions.



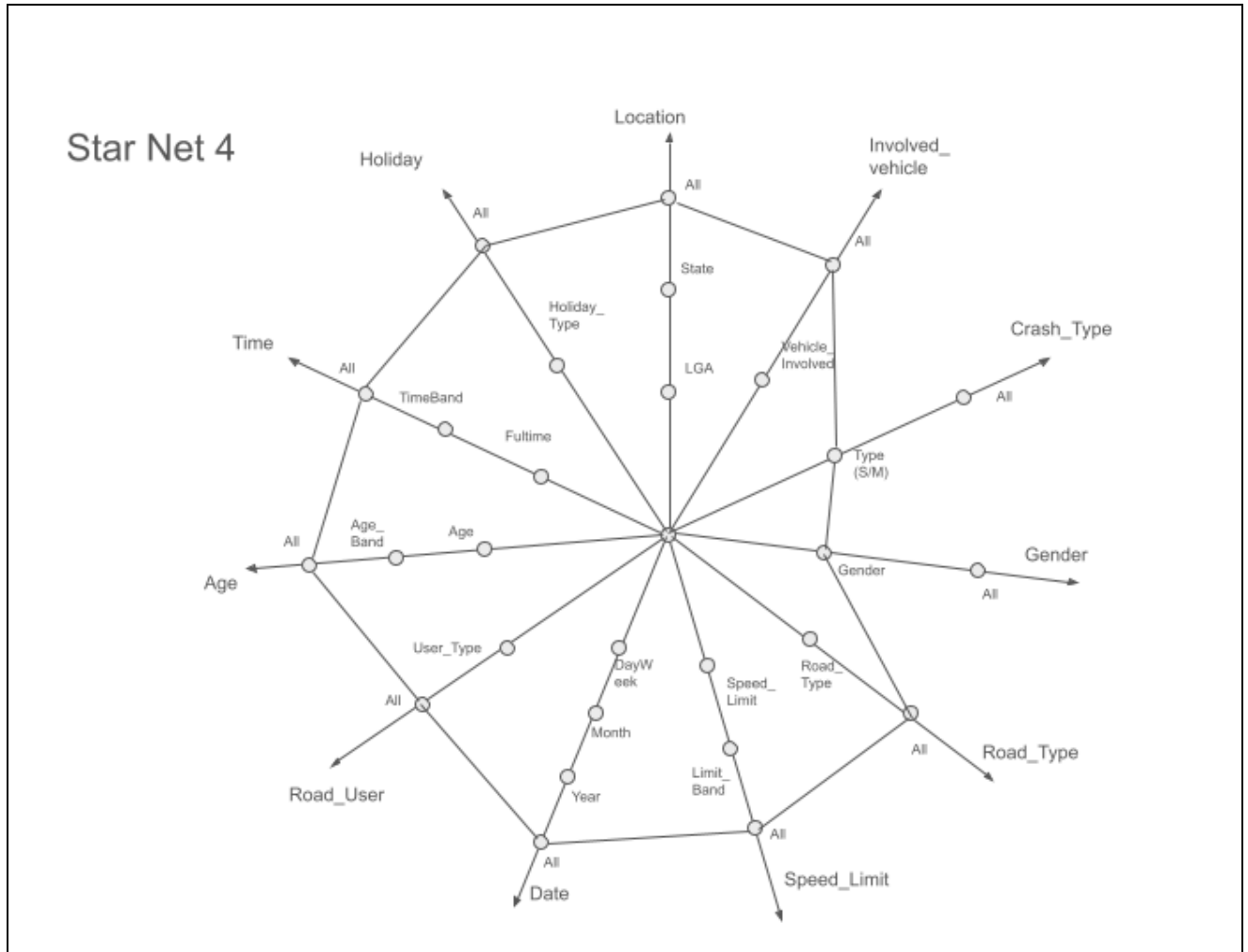
Star Net #2: Number of crashes at night from 2020 for every state. In this star net, our query considers a drill-down to TimeBand for the Time dimension, a drill-down to State for the Location dimension, and a drill-down to Year, for the Date dimension, while rolling up to all other dimensions.



Star Net #3: Number of fatalities on holiday at every speed limit. This star net performs a drill-down to Holiday Type for the Holiday Dimension, a drill-down to Speed\_Limit for the speed limit dimension, and considers the roll-up for all other dimensions.

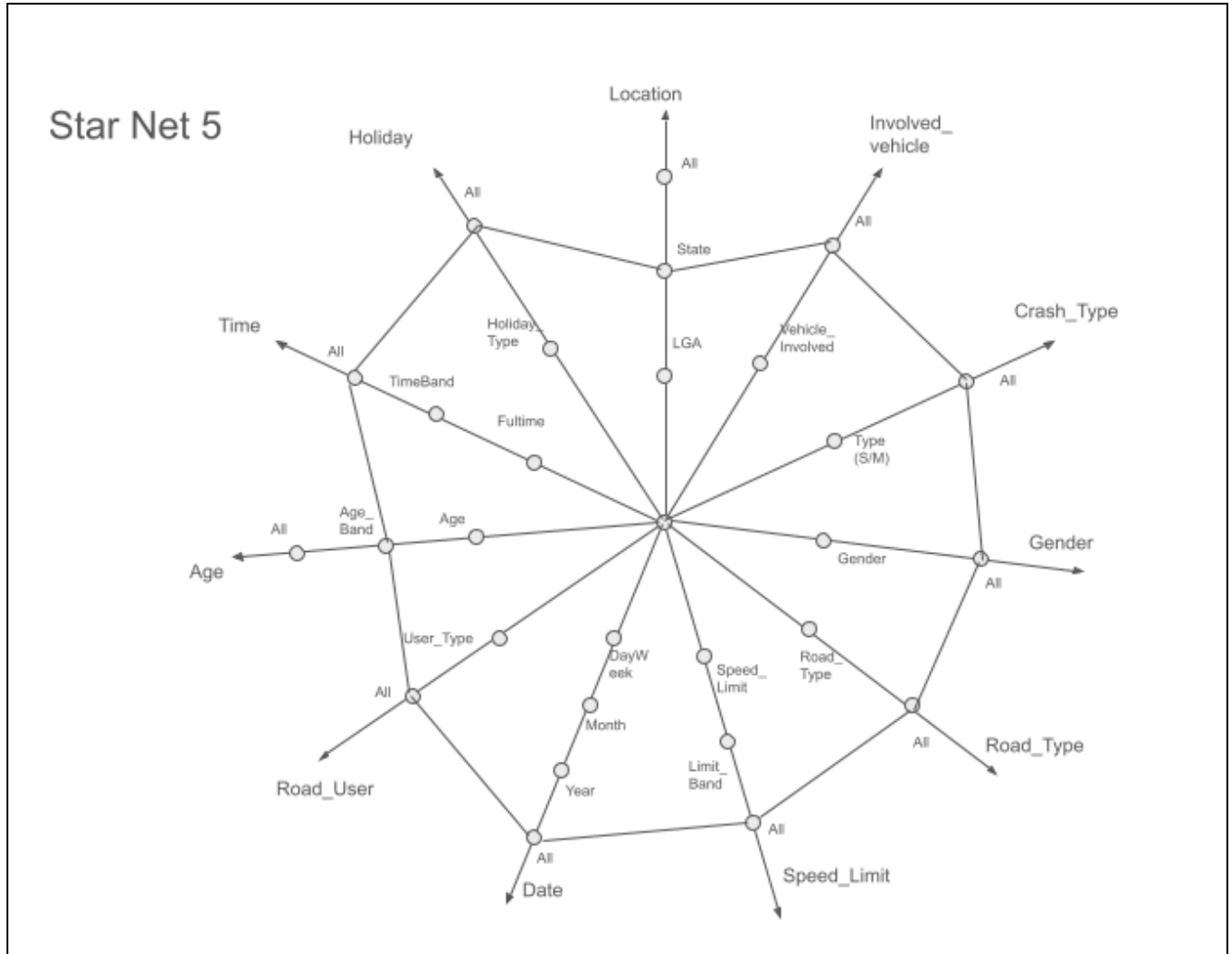


Star Net #4: Percentage of each type of crash by gender. This star net performs a drill-down to Crash\_Type for the Crash\_Type dimension, and a drill down to Gender, for the Gender dimension, while considering the roll-up for all other dimensions.

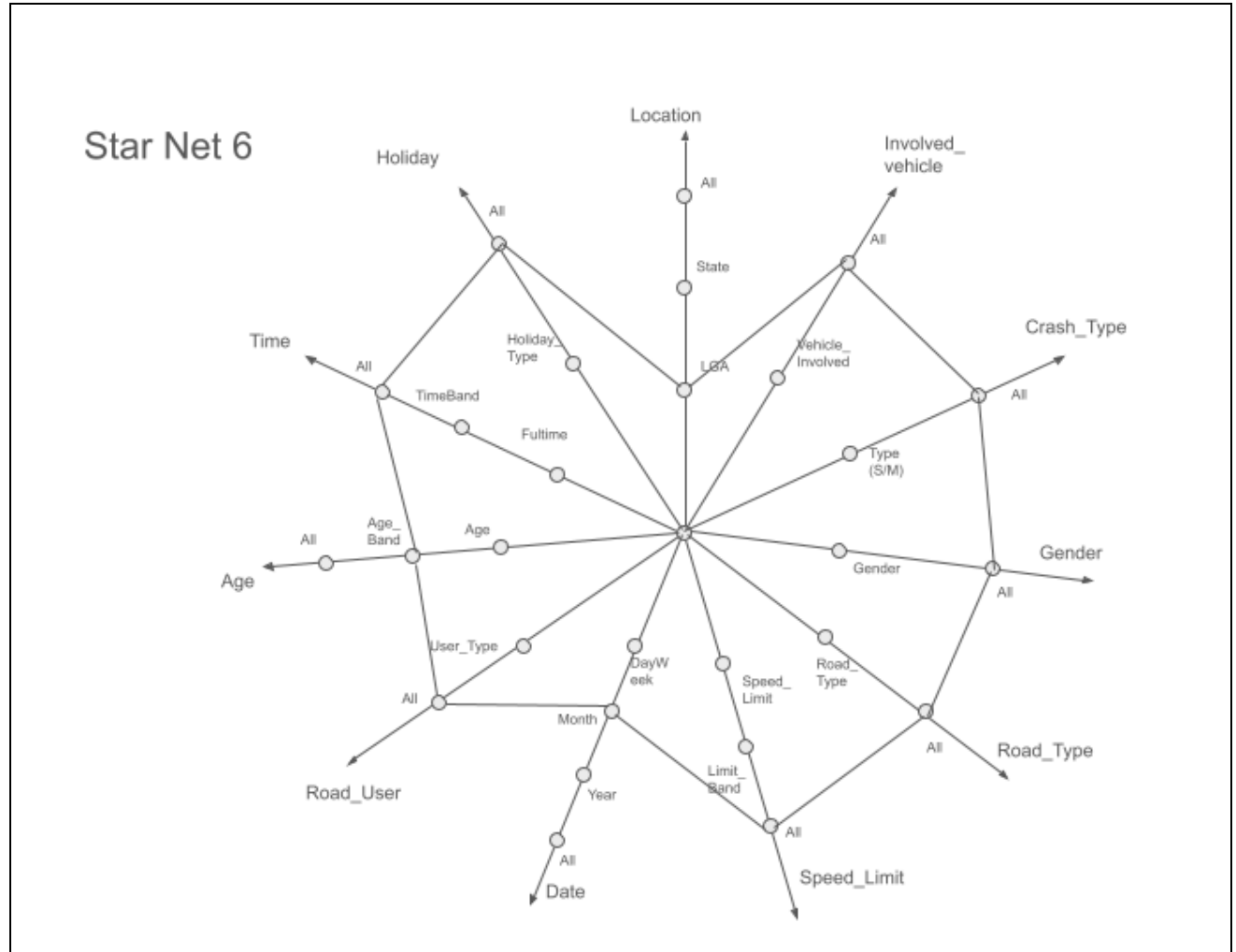




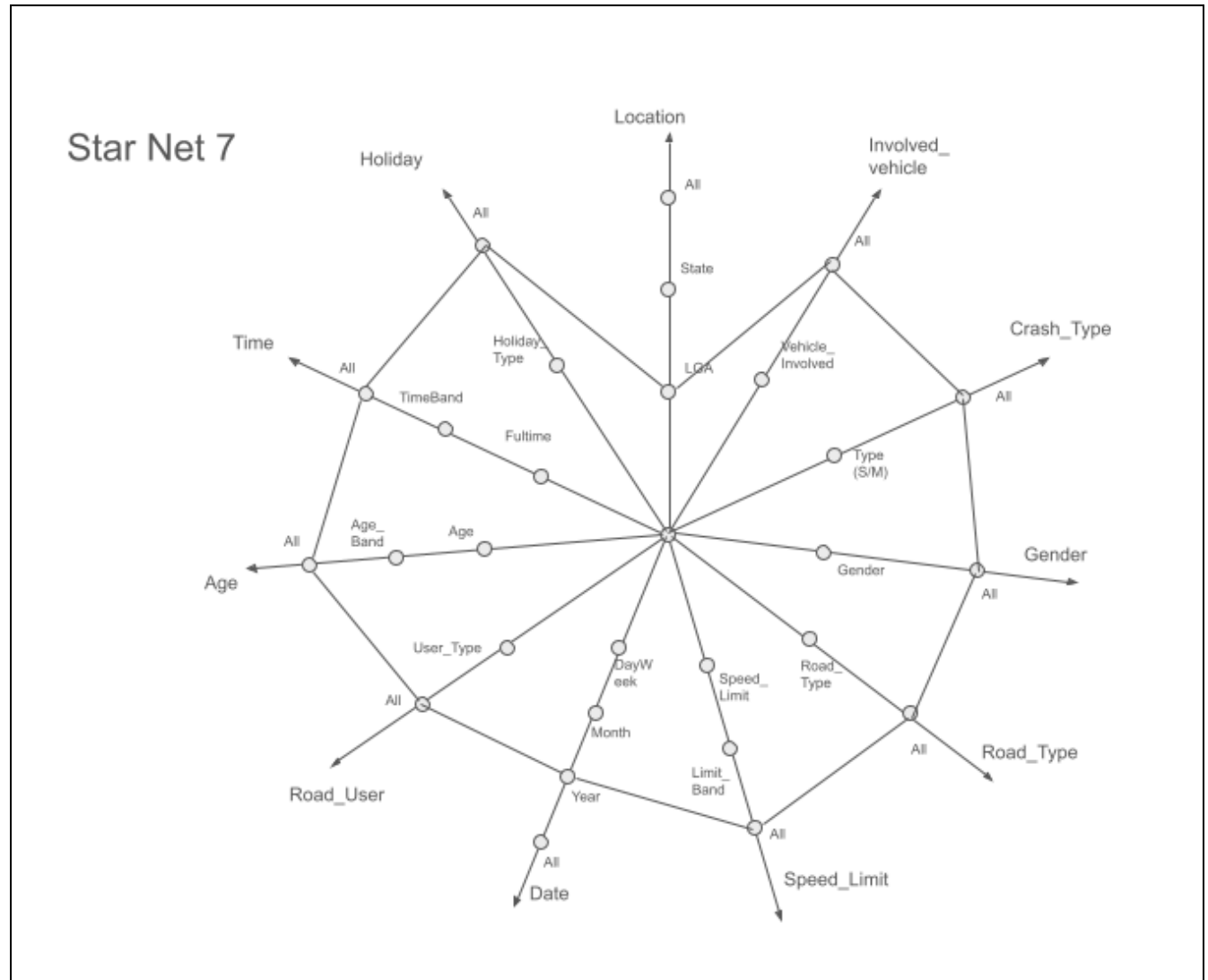
Star Net #5: Number of elderly fatalities in every LGA in WA state. This star net performs a drill-down to State for the Location dimension, and a drill down to Age\_Band, for the Age dimension, while considering the roll-up for all other dimensions.



Star Net #6: Average monthly number of fatalities of young driver in the *Brisbane* LGA. This star net performs a drill-down to Age Band for the Age dimension, a drill-down to Month for the Date dimension, and a drill-down to LGA for the Location dimension, while considering the roll-up for all other dimensions.



Star net #7: Ratio of fatalities over population in Perth in 2024. This star net performs a drill-down to LGA for the Location dimension, and a drill down to Year, for the Date dimension, while considering the roll-up for all other dimensions.



## 2.3. ETL Process

We will discuss the tools and techniques that are used to Extract, Transform, and Load the data to our data warehouse.

### 2.3.1. Extract

To support our warehouse design and analysis, we utilised the crashes and fatalities datasets extracted from Australian Road Deaths Database [2]. Additionally, we extracted demographic data in different geographic area type by the Australian Bureau of Statistics to enrich our analysis and improve the design capability to answer from wider perspectives[6].

Initially, the fatalities dataset consists of 56,874 records of 23 features such as Day, Time, Australian geographical data, Heavy Traffic involvement, Road dimensions, Crash details, and demographic information of road users. The dataset was recorded to each human casualty grain level. However, the dataset does not have its own identification feature.

On the other hand, the crashes dataset consists of 51,284 instances of 20 columns with similar features to fatalities dataset, apart from the road user demographic details. As the data is recorded per crash which is a lower granular level as one crash can result in multiple fatalities, the set has *Number of fatalities* column which presents the total number of fatalities aggregated by Crash ID.

Lastly, although the demographic data consists of several datasets presented by different geographical data, the design only focuses on answering the LGA geographical level. Therefore, only the LGA data was extracted. The LGA demographic dataset contains 547 rows and 25 columns. The populations are presented in a wide format where each row represents the LGA by its code and name and the columns present the year from 2001 to 2023.

### 2.3.2. Transform

The following step is to clean and transform the raw data into the forms and attributes most desirable for the analytics. Although the design objective is to answer queries on fatality level and majority of the features in crash dataset are included in fatality dataset, the first approach is to integrate the two datasets to ensure that all details are aligned. Despite having the same number of crashes and aligning in the time range, the two sets have 2 major mismatched-details issues:

- Crash ID number 20164024 was recorded with 2 fatalities with 2 different Speed Limit. Since one of them was the same with the Crash information and other features are aligned. The Speed Limit of 100 were set back to 60 for consistency.

| Crash ID | State | Month | Year | Dayweek  | Time     | Crash Type | Bus Involvement | Heavy Rigid Truck Involvement | Articulated Truck Involvement | Speed Limit | National Remoteness Areas | SA4 Name 2021    |
|----------|-------|-------|------|----------|----------|------------|-----------------|-------------------------------|-------------------------------|-------------|---------------------------|------------------|
| 20164024 | SA    | 3     | 2016 | Thursday | 08:15:00 | Single     | No              | No                            | No                            | 60          | Major Cities of Australia | Adelaide - North |
| 20164024 | SA    | 3     | 2016 | Thursday | 08:15:00 | Single     | No              | No                            | No                            | 100         | Major Cities of Australia | Adelaide - North |

- There are 275 instances of National Road Type that were recorded in Crash dataset but were unavailable in Fatality dataset. The missing data was filled in the Fatality dataset for consolidation.

Since we had ensured that the crash data extracted from the Fatality dataset and the Crash data itself are perfectly aligned, we use the Fatality dataset as a starting point to extract and transform the fact and dimension tables into csv format.

Prior to each dimension extraction, data cleaning is performed to deal with missing data. A threshold of 5% of the dataset, which is roughly 2,800 rows, was set to avoid bias and unrepresentativeness of the data. 1,691 rows with missing features were removed. However, the majority of instances were still missing LGA geographical, heavy rigid truck involvement, and road type details. The missing LGA data is imputed with "Unidentified\_State" where State is the corresponding state of the instances. On the other hand, the missing data of Heavy Rigid Truck Involvement and National Road Type is filled with "Unknown\_Involvement" and "Unknown\_roadtype", respectively.

```
fatalities_clean['National LGA Name 2021'] = fatalities_clean['National
LGA Name 2021'].fillna(value = 'Unidentified_' +
fatalities_clean['State'])
fatalities_clean['National Road Type'] =fatalities_clean['National Road
Type']
.fillna('Unknown_roadtype')
fatalities_clean['Heavy Rigid Truck Involvement'] =
fatalities_clean['Heavy Rigid Truck
Involvement'].fillna('Unknown_Involvement')
```

Dimension tables are extracted by normalisation, feature selection and dimension reduction. We subsetting fatality data into multiple data frames which only have the features following the design and would be normalised to only keep unique values. Features that are not aligned with the design would be dropped at this stage.

However, population dimension was handled separately. The population geographical dataset is clean so minimal data cleaning was required. However, since the LGA name in fatality dataset is 2021 version while the population dataset is 2023, there are few mismatches with the fatality dataset:

- Moreland LGA region updated its name to Merri-bek

- Fatality dataset has inconsistency in naming region with similar name but different state such that region not belonged to NSW have the format “LGA (State)” while region within NSW only has “LGA”
- Unincorporated areas were recorded in different styles on both file

The LGA names in Location Dimension were updated according to the population dataset. The population dataset was then converted to long format with each instance corresponding to the population of a LGA area at a time. LGA name and code were replaced with location\_id for it to link back to location dimension.

Additionally, a GeoJson file containing the geographic data of each LGA area in multipolygon format was used to support map visualisation. However, as the dataset originated from a different source, discrepancies were unavoidable. Since Tableau built-in function allows direct import of geographical data, the dataset is not included in the data warehouse design. Our only approach to remedy the discrepancies is to introduce the LGA\_NAME21 in location dimension which will allow Tableau to correctly recognise the area by name and visualise the data by LGA.

### 2.3.3. Load

The dimension and facts are uploaded to the data warehouse using a SQL script. Consequently, the data will be available for analysis and query purposes. Visualisation tools such as Tableau can connect directly to the OLAP server and retrieve the data for generating graphs and charts.

```
COPY dim_date FROM '/tmp/Assignment_1/Dataset/date_dimension.csv' WITH (FORMAT csv, HEADER);
COPY dim_time FROM '/tmp/Assignment_1/Dataset/time_dimension.csv' WITH (FORMAT csv, HEADER);
COPY dim_roaduser FROM '/tmp/Assignment_1/Dataset/road_user_dimension.csv' WITH (FORMAT csv, HEADER);
COPY dim_gender FROM '/tmp/Assignment_1/Dataset/gender_dimension.csv' WITH (FORMAT csv, HEADER);
COPY dim_age FROM '/tmp/Assignment_1/Dataset/age_dimension.csv' WITH (FORMAT csv, HEADER);
COPY dim_road FROM '/tmp/Assignment_1/Dataset/road_dimension.csv' WITH (FORMAT csv, HEADER);
COPY dim_speed FROM '/tmp/Assignment_1/Dataset/speed_dimension.csv' WITH (FORMAT csv, HEADER);
COPY dim_crashtype FROM '/tmp/Assignment_1/Dataset/crash_dimension.csv' WITH (FORMAT csv, HEADER);
COPY dim_involvedtype FROM '/tmp/Assignment_1/Dataset/involvement_dimension.csv' WITH (FORMAT csv, HEADER);
COPY dim_location FROM '/tmp/Assignment_1/Dataset/location_dimension.csv' WITH (FORMAT csv, HEADER);
COPY dim_holiday FROM '/tmp/Assignment_1/Dataset/holiday_dimension.csv' WITH (FORMAT csv, HEADER);
COPY dim_population FROM '/tmp/Assignment_1/Dataset/population_dimension.csv' WITH (FORMAT csv, HEADER);
COPY fact_fatalities FROM '/tmp/Assignment_1/Dataset/fact_fatalities.csv' WITH (FORMAT csv, HEADER);
```

## 2.4. Implementation

We have explored the data using the PostgreSQL interface on pgAdmin to find the answers to the business questions. In addition, we explored the data visually using Tableau,

SQL Queries:

1. What is the number of fatalities at different times of the day?

--Question 1: What is the number of fatalities at different times of the day?

```
SELECT t.timeband, COUNT(*) AS total_fatalities
FROM fact_fatalities AS f
LEFT JOIN dim_time AS t ON t.time_id = f.time_id
GROUP BY CUBE (t.timeband)
ORDER BY t.timeband;
```

| TimeBand  | Fatalities |
|-----------|------------|
| Afternoon | 15,530     |
| Evening   | 11,417     |
| Morning   | 14,235     |
| Night     | 14,001     |
| All       | 55,183     |

Table 2.4.1. Number of fatalities at different time

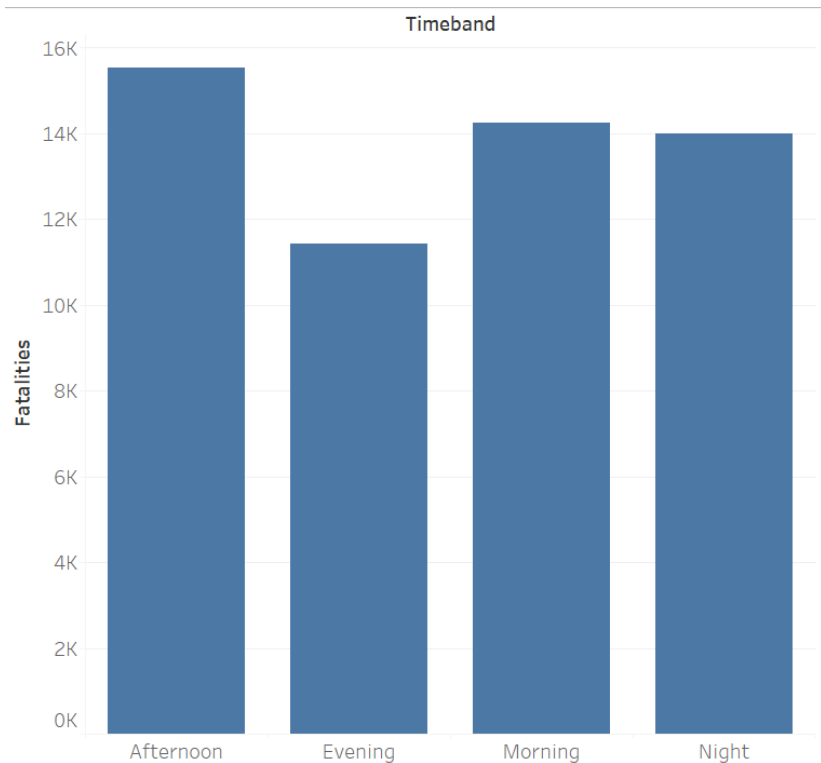


Fig 2.4.2. Number of fatalities at different time

Afternoon period has the most number of fatalities, while the evening has the least. It's potential that afternoon corresponds to the time people leave their exhausted workday and rush back to their homes. Driving in fatigue condition and noncompliance to speed limit poses high risk to traffic accident.

2. What is the number of crashes at night from 2020 of every states?

--Question 2: What is the number of crashes at night from 2020 of every states?

```
SELECT l.state, COUNT(DISTINCT f.crash_id) AS total_crashes
FROM fact_fatalities AS f
LEFT JOIN dim_time AS t ON t.time_id = f.time_id
LEFT JOIN dim_date AS d ON d.date_id = f.date_id
LEFT JOIN dim_location AS l ON l.location_id = f.location_id
WHERE t.timeband = 'Night' AND d.year = 2020
GROUP BY CUBE(l.state)
ORDER BY total_crashes DESC;
```

| State | Total Crashes |
|-------|---------------|
| All   | 224           |
| NSW   | 59            |
| Qld   | 53            |
| Vic   | 50            |
| WA    | 24            |
| SA    | 15            |
| NT    | 14            |
| Tas   | 8             |
| ACT   | 1             |

Table 2.4.3. Number of Crashes at Night by State in 2020



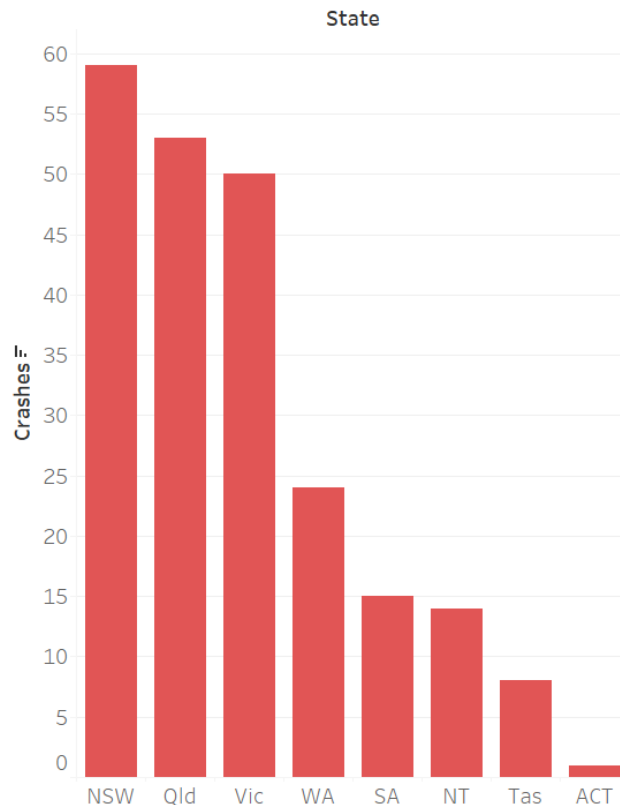


Fig 2.4.4. Number of Crashes at Night by State in 2020

The number of crashes at night is the most in NSW with the most cases while there was only 1 case recorded in ACT. The top 3 states are more than double the following one. However, it might be misleading since populations there are most crowded in Australia. Nevertheless, more educational campaigns have to be carried out to reinforce the awareness of safety driving at night in the 3 states.

3. What is the number of fatalities on holiday at every speed limit?

```
-- Question #3: Number of fatalities on holiday at every speed limit
SELECT ds.speedlimit, COUNT(*) AS fatality_count
FROM fact_fatalities ff
JOIN dim_speed ds ON ff.speedlimit = ds.speedlimit
JOIN dim_holiday dh ON ff.holiday_id = dh.holiday_id
WHERE dh.holiday IN ('Christmas', 'Easter')
GROUP BY ds.speedlimit
ORDER BY ds.speedlimit;
```

| SpeedLimit | Fatalities |
|------------|------------|
| 20         | 2          |
| 40         | 12         |
| 50         | 155        |
| 60         | 418        |
| 70         | 93         |
| 75         | 3          |
| 80         | 224        |
| 90         | 36         |
| 100        | 807        |
| 110        | 282        |
| 130        | 4          |
| All        | 2036       |

Table 2.4.5. Number of fatalities on holiday by speed limit

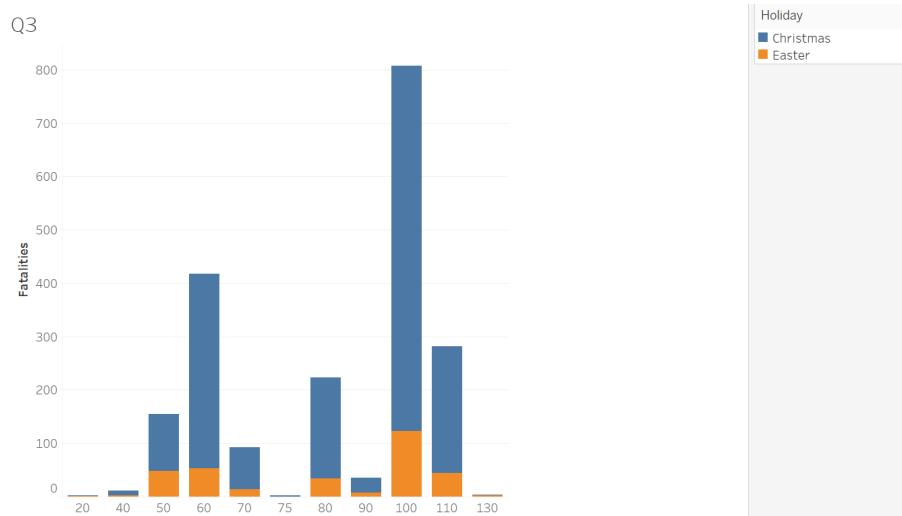


Fig 2.4.6. Barchart Number of fatalities on holiday by speed limit

The number of fatalities distributed in a multimodal distribution where its peak at 60km/h and 100km/h. However, the graph explicitly demonstrates that there are more fatalities at higher speed roads. Additionally, the number fatalities during Christmas period is significantly higher than Easter. The alarming fact is urging for deeper investigation in the matter while implement more campaigns during this holiday.

4. What is the percentage of each type of crash by gender?

```
--Question 4: What is the percentage of each type of crash by gender?
SELECT ct.typelabel, g.gender,
       ROUND(100.0 * COUNT(DISTINCT f.crash_id) / SUM(COUNT(DISTINCT f.crash_id)) OVER
         (PARTITION BY g.gender), 2) AS percentage
FROM fact_fatalities AS f
LEFT JOIN dim_gender AS g ON g.gender_id = f.gender_id
LEFT JOIN dim_crashtype AS ct ON ct.crashtype_id = f.crashtype_id
GROUP BY ct.typelabel, g.gender
ORDER BY ct.typelabel, g.gender
```

| Crash Type | Gender | Percentage |
|------------|--------|------------|
| Multiple   | Female | 48.8       |
| Multiple   | Male   | 41.82      |
| Single     | Female | 51.2       |
| Single     | Male   | 58.18      |

Table 2.4.7 Table Percentage of each type of Crash by Gender

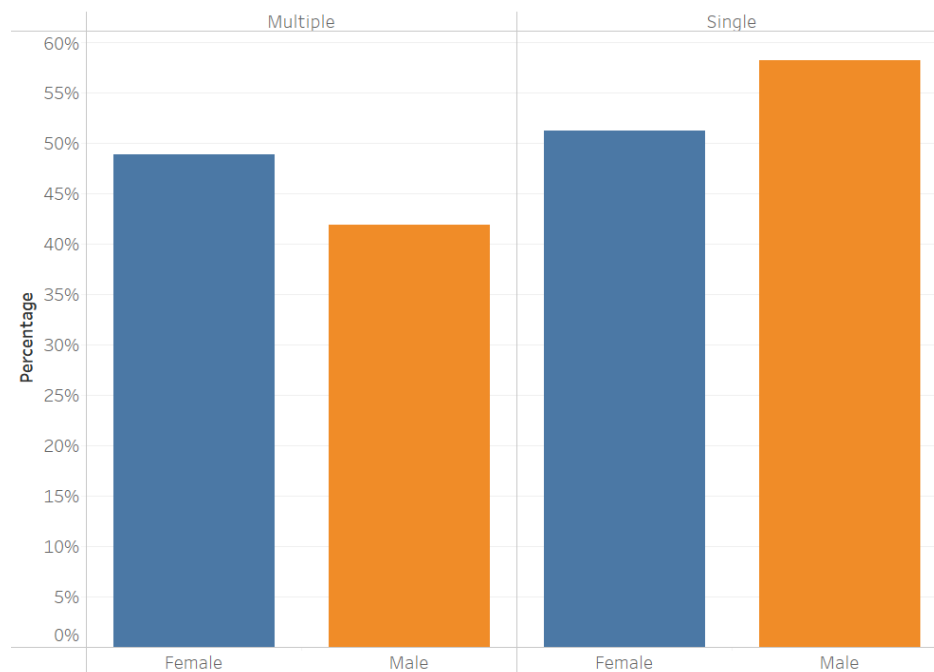


Fig 2.4.7 Barchart Percentage of each type of Crash by Gender

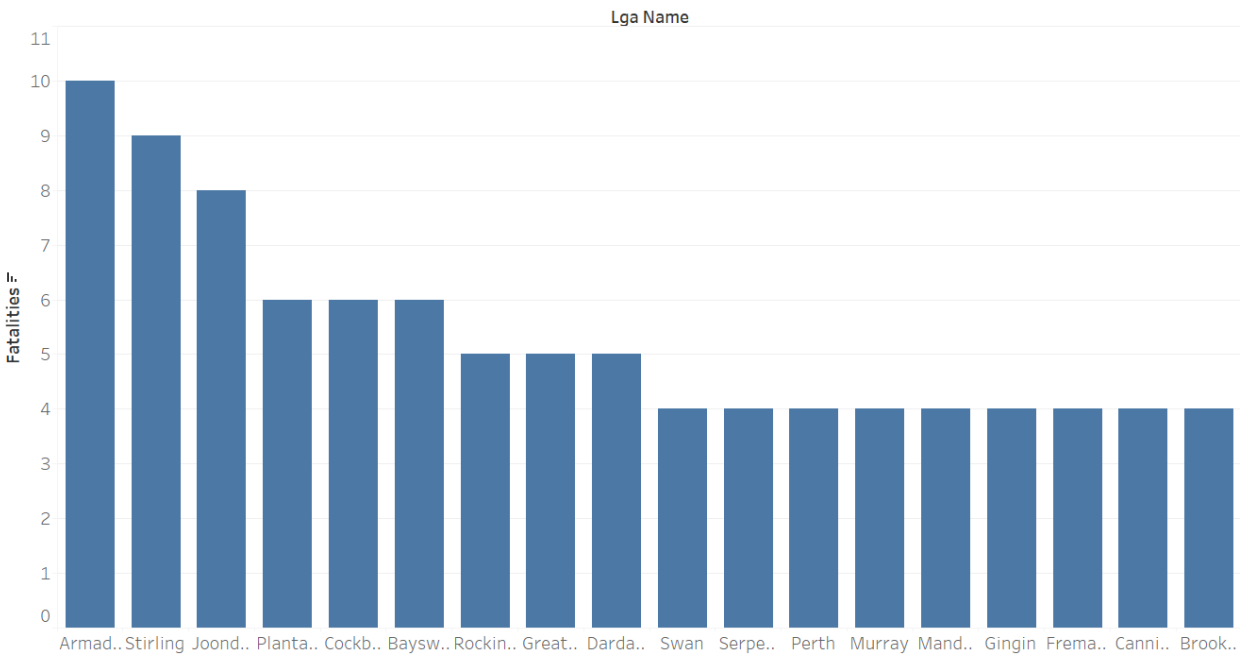
The fact and figure demonstrate that multiple crashes is more likely to be caused by female while single crashes is more likely to be caused by male. However, no explicit implication can be drawn from this. As the query and chart is to demonstrate the capability of the warehouse on handling multi dimensional data problem.

5. What is the top 10 number of elderly fatalities in every LGA in WA state?

```
--Question 5: What is the top 10 number of elderly fatalities in every LGA in WA state?  
SELECT l.lga_name, COUNT(*) AS elderly_fatalities  
FROM fact_fatalities AS f  
LEFT JOIN dim_age AS a ON a.age = f.age  
LEFT JOIN dim_location AS l ON l.location_id = f.location_id  
WHERE a.ageband = 'Older' AND l.state = 'WA'  
GROUP BY l.lga_name  
ORDER BY elderly_fatalities DESC  
LIMIT 10;
```

| LGA               | Elderly Fatalities |
|-------------------|--------------------|
| Unidentified_WA   | 704                |
| Armadale          | 10                 |
| Stirling          | 9                  |
| Joondalup         | 8                  |
| Plantagenet       | 6                  |
| Bayswater         | 6                  |
| Cockburn          | 6                  |
| Rockingham        | 5                  |
| Greater Geraldton | 5                  |
| Dardanup          | 5                  |

Table 2.4.8. Top 10 LGA in WA by Elderly Fatalities



#### Fig 2.4.8. Barchart Top 10 LGA in WA by Elderly Fatalities

The table illustrates that the majority of the fatalities at elderly people were not identified on LGA level. The barchart was modified to remove the unidentified area to present a less skewed visual. Due to the lack of quality in data, it is too risky to draw any insight from the presentation. However, the design is proved to be capable of answering the business question but requiring more investment in the data integrity.

6. What is the average monthly number of fatalities of young driver in the *Brisbane* LGA in last 5 years?

```
--Question 6: What is the average monthly number of fatalities of young driver in the Brisbane LGA in 2023?
SELECT d.year, ROUND(COUNT(*)*1.0/12,2) AS monthly_young_fatalities
FROM fact_fatalities AS f
LEFT JOIN dim_age AS a ON a.age = f.age
LEFT JOIN dim_location AS l ON l.location_id = f.location_id
LEFT JOIN dim_date AS d ON d.date_id = f.date_id
WHERE d.year BETWEEN 2020 AND 2024
      AND l.lga_name = 'Brisbane'
      AND a.ageband = 'Young'
GROUP BY d.year
ORDER BY d.year
```

| Year | Monthly Young Fatalities |
|------|--------------------------|
| 2020 | 0.33                     |
| 2021 | 0.5                      |
| 2022 | 0.42                     |
| 2023 | 0.67                     |
| 2024 | 0.75                     |

Table 2.4.9. Monthly young fatalities in Brisbane last 5 years

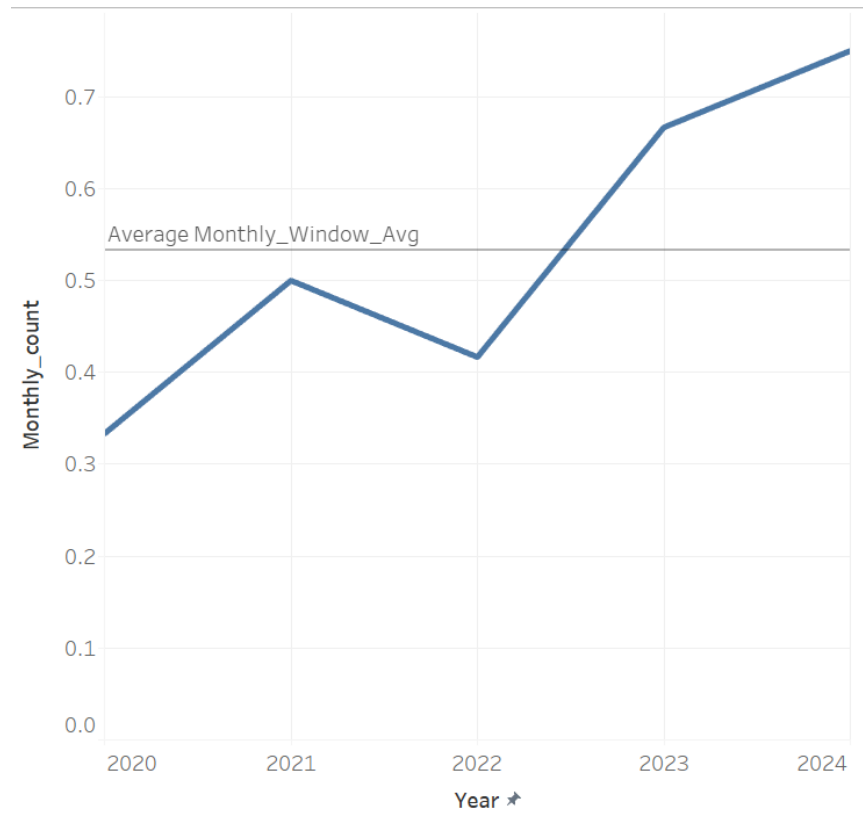


Fig 2.4.10. Linechart monthly young fatalities in Brisbane last 5 years

The table and line chart illustrates that the monthly fatalities among young road users in Brisbane has gradually increased over the last 5 years. The authority is recommended to carry out educational marketing campaigns on social media platforms which are popular amongst young people. The solution can help raise the awareness of safety participating in the youth. Although, the number would likely to increase but the trend is unlikely to remain as the majority of LGA data are missing.

7. What is the ratio of fatalities in 100,000 people in Perth in 2024?

--Question 7: What is the ratio of fatalities in 100,000 people in Perth in 2024?

```
WITH perth_fatalities AS (
  SELECT lga_name, year, COUNT(*) AS total_fatalities
  FROM fact_fatalities AS f
  LEFT JOIN dim_location AS l ON l.location_id = f.location_id
  LEFT JOIN dim_date AS d ON d.date_id = f.date_id
  WHERE year >= 2020 AND lga_name = 'Perth'
  GROUP BY lga_name, year
),
perth_population AS(
  SELECT lga_name, year, population
  FROM dim_population AS p
  LEFT JOIN dim_location AS l ON p.location_id = l.location_id
  WHERE lga_name = 'Perth' AND year >= 2020
)
SELECT f.lga_name, f.year, total_fatalities, population,
       total_fatalities*100000.0/population AS ratio
FROM perth_fatalities AS f
LEFT JOIN perth_population AS p ON f.lga_name = p.lga_name AND f.year = p.year
```

| LGA   | Year | Total Fatalities | Population | Ratio       |
|-------|------|------------------|------------|-------------|
| Perth | 2020 | 1                | 29588      | 3.379748547 |
| Perth | 2021 | 1                | 29653      | 3.372340067 |

Table 2.4.11. Ratio of fatalities over 100,000 people in Perth last 5 years

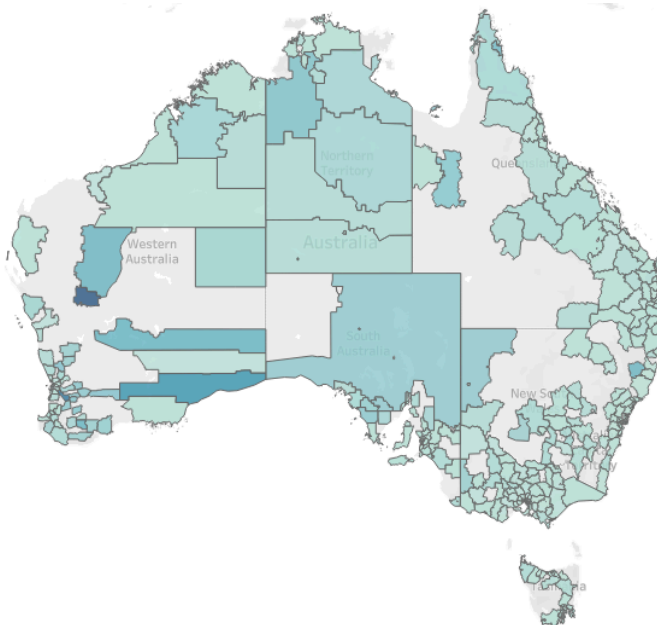


Fig 2.4.12. Heatmap of ratio of fatalities over 100,000 people per LGA per year

Perth only recorded 1 fatalities each year in 2020 and 2021 and there was no case recorded in the last 3 years. This can be either due to the missing LGA data or the data is actually true since Perth LGA is an urban central area where vehicles can only travel relatively slowly. The map demonstrates the ratio by LGA where the darker color can be translated to severity. Though the integration of Tableau with our design is capable of producing meaningful visualisations, the poor quality in data restrained its from being on LGA level.

## 2.5. Association rule mining

Association Rule Mining is a powerful tool that can be applied to identify interesting patterns, associations and correlations amongst the items in large dataset. In this report, the Apriori algorithm was employed to investigate the underlying relationships of factors in fatalities.

The Apriori has been successfully applied in wide range of problems such as market basket analysis, and similar road accident analysis [7], [8]. Apriori operates on the principle that if an itemset is frequent, hence, its subsets must also be frequent. Therefore, the algorithm systematically generates candidate itemsets and prunes those below the set support threshold. Significant patterns can be extracted differently by employing different ranking rules. This offers various perspectives to the problem, facilitating policymakers to tailor the solutions suiting their region. Furthermore, the mlxtend package allows seamless implementation of Apriori in Python [9].

Regarding the analysis, the goal is to draw insights of factors that influence the fatality of road users. Hence, we employ the top k rules that have “Road User” on the right-hand side. Intuitively, top k represents the number k, where  $k \geq 1$ , of top performers regarding the ranking metrics. On the other hand, in association rule mining, the rule takes the form of  $A \rightarrow B$  where A represents the Antecedent and B represents the Consequent. In this case, the “Road User” on the right-hand side is referred to “Road User” as the Consequent of the rules mined. The performance will be ranked by confidence and lift separately where confidence refers to the probability of type of “Road User” occurrence given A and lift refers to the increasing likelihood that type of “Road User” appears when A is present.

As discussed in the previous section, the majority data of Heavy traffic involvement and LGA name and type of road are missing. Furthermore, Apriori is only efficient at handling categorical data. Heavily skewed missing data and numerical data are not appropriate for this type of analysis. Therefore, the analysis focuses on drawing the insights of human behaviours from demographic data such as gender, age group, environmental factors such as time, speed limit band, and state geographical level for policy tailored. However, in this particular analysis, we introduced a Weekend Categorical dimension where “Weekend” value refers to “Sunday” and “Saturday” while the rest are “Weekday”. A support threshold of 5%, which is roughly 2,750 similar instances, was set to prune sets of items that have the frequency below 5%. Additionally, the confidence threshold of 50% was set to only focus on “strong” association where the consequent has to occur at least 50% of the time the antecedent happens.

The tables below present the result of top 5 rules ranked by confidence and lift.



| antecedents                | consequents | confidence |
|----------------------------|-------------|------------|
| (MidAge, Weekday, Over100) | (Driver)    | 0.66193    |
| (MidAge, Male, Over100)    | (Driver)    | 0.641057   |
| (Weekday, Male, Over100)   | (Driver)    | 0.634286   |
| (Male, Over100, Morning)   | (Driver)    | 0.633198   |
| (Male, Over100, Night)     | (Driver)    | 0.629083   |

Table 2.5.1. Top 5 rules ranked by Confidence

(MidAge, Weekday, Over100) -> Driver: If the fatality is a person in their Middle Age on Weekday in a road where speed limit is over 100, then there is a 66.2% chance that person is a driver

(MidAge, Male, Over100) -> Driver: If the fatality is male middle-aged person on a over 100 km/h speed limit road, then there is 64.1% that person is a driver

(Weekday, Male, Over100) -> Driver: If the fatality is a male person during weekday on a over 100 speed limit road, then there is 63.4% that person is a driver

(Male, Over100, Morning) -> Driver: If the fatality is a male person in the morning on a over 100 speed limit road, then there is 63.3% that person is a driver

(Male, Over100, Night) -> Driver: If the fatality is a male person at night on a over 100 speed limit road, then there is 62.9% chance that person is a driver

| antecedents                | consequents    | Lift     |
|----------------------------|----------------|----------|
| (Over100, Night)           | (Male, Driver) | 1.488788 |
| (MidAge, Weekday, Over100) | (Driver)       | 1.45771  |
| (MidAge, Male, Over100)    | (Driver)       | 1.411743 |
| (Weekday, Male, Over100)   | (Driver)       | 1.396831 |
| (Male, Over100, Morning)   | (Driver)       | 1.394436 |

Table 2.5.2. Top 5 rules ranked by Lift

(Over100, Night) -> (Over100, Night): A fatality happened on a over 100 km/h speed limit road at night is 1.49 times more likely to be a male driver compared to the antecedent and the consequent happening independently

(MidAge, Weekday, Over100) -> Driver: A fatality of a middle-aged person during weekday on a over 100 speed limit road is 1.46 times more likely to be a driver compared to the antecedent and the consequent happening independently

(MidAge, Male, Over100) -> Driver: A fatality of a male middle-aged person on a over 100 speed limit road is 1.41 times more likely to be a driver compared to the antecedent and the consequent happening independently

(Weekday, Male, Over100) -> Driver: A fatality of a male person during weekday on a over 100 speed limit road is 1.40 times more likely to be a driver compared to the antecedent and the consequent happening independently

(Male, Over100, Morning) -> Driver: A fatality of a male person in the morning on a over 100 speed limit road is 1.49 times more likely to be a driver compared to the antecedent and the consequent happening independently

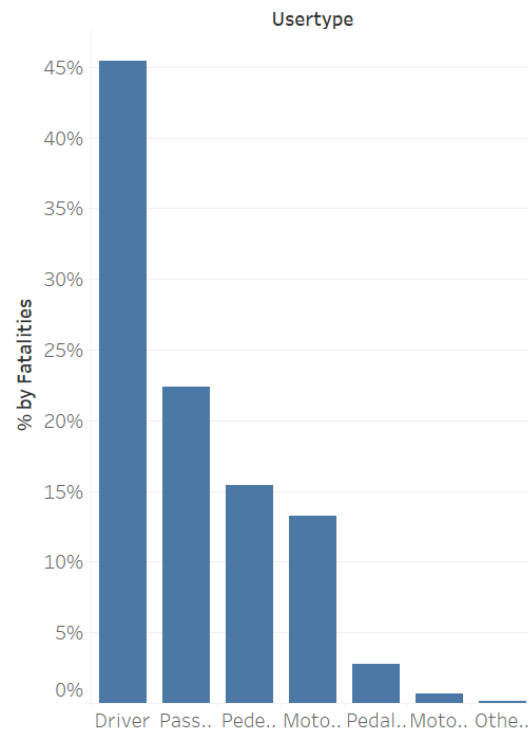


Fig 2.5.3. Barchart of percentage by fatalities amongst Road User

The antecedents point toward the Driver is reasonable as the data collected heavily skewed to the Driver along with the confidence threshold of 50% eliminates less likely occurrence Road User. Several key takeaways can be drawn from the mined result. Male drivers are consistently involved in high-speed crashes. Weekdays on high-speed roads pose more risks to drivers, potentially because driving fast in a crowded traffic tends to lead to accidents. Driving at night on high-speed roads is particularly dangerous for male drivers. Though these cases can potentially be related to driving after drinking, insufficient light system on the road, overspeeding since little of road traffic, hence, more investigation and data collected should be invested to find the exact root causes.

Since the driver is the most important element when controlling the vehicle and their behaviour can dictate the outcome, the authorities should invest more in educational marketing campaigns to raise awareness and reinforce the road users' sense of responsibility. Furthermore, crashes during weekdays are likely to be related to work exhaustion, heavy traffic, stress and pressure. Therefore, a dynamic speed limit system on high-speed roads should be implemented during weekdays to reduce the chance of accidents. More signs should be put on the high speed roads

to remind the drivers of potential danger. Increasing speed and fatigue checkpoint is promising to reduce the driver's chance of engaging in speeding and fatigue driving, hence minimising the risks of accidents.

### 3. Conclusion

This study demonstrated the application of Data Warehousing and Association Rule Mining in Road Safety Information Management. The data warehouse framework was designed to store historical data of traffic fatalities along with their related features while following the OLAP data warehouse model. The implementation of the design showcased its capability to handle multi-dimensional queries efficiently, facilitating a fast and accurate decision-making process. Additionally, BI applications can leverage the warehouse to retrieve and visualise data, further refining the process and significantly improving the quality of the decision.

Despite the effectiveness, there are few drawbacks within the design, primarily the concern of data quality. Incomplete records highlight the need for improvement in data collection and integration. Furthermore, the future research should explore additional factors potentially influencing the fatality such as alcohol consumption, weather features, and road conditions, to enhance the analysis and develop more comprehensive safety implications.

On the other hand, the application of Apriori algorithm helped uncover insightful patterns in fatal road accidents. By mining historical data, key risk factors associated with road users, particularly the driver, were indicated.

The results from both analyses implied an intervention from the authority on high-speed roads. The findings suggest that a dynamic speed control system should be implemented to minimise the chance of crash. Additionally, targeted educational campaigns through multiple platforms can reinforce awareness and encourage compliance with traffic regulations, consequently reducing risk of fatal accidents.

Overall, this study highlights the strength of Data Warehousing and Association Rule Mining in road safety data management, analysis, and policy-making implication. With proper implementation, the technologies can enhance organisational decision-making and contribute significantly to the success.

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